H.T.No: | | | | | | | | | | |

Course Code No: 23IT3T01

VISHNU INSTITUTE OF TECHNOLOGY (AUTONOMOUS)

II B. Tech I Semester (R23) Regular Examinations, November– 2024

OBJECT ORIENTED PROGRAMMING THROUGH JAVA

(CSE, IT, AI & DS , AI & ML , CS & BS)

Time:3hours **Max. Marks: 70M**

Note:1. Question Paper consists of two parts (**Part-A** and **Part-B**)
2. Answer all the questions in **Part-A**
3. Answer either a or b questions from Question number 2 to 6 in **Part-B**

PART –A

1	I	Define the principles of object-oriented programming.	L1	CO1	[2M]
	II	Write the syntax for declaring a method in Java.	L2	CO2	[2M]
	III	Differentiate between inheritance and interface in Java.	L1	CO3	[2M]
	IV	Explain the use of the super keyword.	L1	CO4	[2M]
	V	What is an abstract class? Provide an example.	L1	CO5	[2M]
	VI	List and explain any four types of operators in Java	L1	CO1	[2M]
	VII	Define method overloading.	L1	CO2	[2M]
	VIII	Write any two advantages of using packages in Java	L1	CO3	[2M]
	IX	What is exception handling? Mention its key components.	L1	CO4	[2M]
	X	List and explain two methods from the Thread class.	L1	CO5	[2M]

PART –B

UNIT – I

2(a)		List and explain java features in detail.	L2	CO1	[10 M]
		(OR)			
2(b)	I	Write a program to check whether the given number is palindrome or not.	L2	CO1	[5 M]
	II	Explain the usage of for loop with an example.	L2	CO1	[5 M]

UNIT – II

3(a)	I	Define constructors in Java. Explain constructor overloading with an example program..	L2	CO2	[6 M]
	II	Explain the concept of nested classes in Java. Write a program to demonstrate their usage	L2	CO2	[4 M]

(OR)

3(b)	I	*Write a program to demonstrate method overloading.*	L3	CO2	[5M]
	II	*Write a program to demonstrate the use of this keyword to refer to current class variables and methods*	L2	CO2	[5 M]
UNIT – III					
4(a)	I	Explain the process of inheritance in Java with a detailed example	L3	CO3	[5M]
	II	Explain the process of dynamic method dispatch in Java with a program.	L2	CO3	[5M]
		(OR)			
4(b)		*Write a program to perform matrix addition using two-dimensional arrays in Java.*	L2	CO3	[10M
UNIT – IV					
5(a)	I	Explain how to handle multiple exceptions in java. Write a program to handle divide-by-zero, null-pointer, and array-index-out-of-bound exceptions	L2	CO4	[6M
	II	*Write a program to write data to a file. Explain the methods used*	L3	CO4	[4M
		(OR)			
5(b)		Discuss the concept of Java packages. Write an example to demonstrate creating and importing a user-defined package	L2	CO4	[10M
UNIT – V					
6 (a)	I	.Explain the thread life cycle in Java with a neat diagram and examples	L2	CO5	[5 M
	II	Write a program to create multiple threads by implementing Runnable interface.	L3	CO5	[5 M
		(OR)			
6(b)		Write a program to connect to a database using JDBC, retrieve data, and display it.	L2	CO5	[10M

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